

Course 6031D

Electric Vehicles

Diploma Engineering Handbook Notes

POLY PMNA - Polytechnic Study Hub

Course 6031D - Electric Vehicles

Semester 6 | Program Elective | Credits: 4 | Periods: 60

Downloadable handbook notes generated from lessons-6031D.html.

Core areas: EV classifications, drive trains, motors, energy storage, EVSE, renewable charging, grid support and maintenance.

Overview and Roadmap

Objectives: express EV relevance for future transportation, understand motor applications in EV, summarize energy storage options, and identify charging infrastructure requirements.

CO1: Outline EV basic concepts and classifications. CO2: Identify EV and HEV drive trains. CO3: Outline hybridization of energy storage. CO4: Explain EV charging systems integrated with renewable energy.

Module roadmap: M1 EV basics, FAME policies and motor drive technologies. M2 EV/HEV drive trains, power flow and regenerative braking. M3 batteries, BMS, converters, EVSE and smart charging. M4 grid integration, V2X, renewable charging, maintenance and recycling.

Module Chapters

Module 1: EVs use electric propulsion and energy storage instead of depending only on internal combustion. Types include BEV, FCEV, HEV and PHEV. Benefits include lower tail-pipe emission, high efficiency, less noise and regenerative braking. Motor drives include DC, induction, permanent magnet and switched reluctance drives.

Module 2: EV subsystems include electric propulsion unit, energy source subsystem and auxiliary subsystem. Drive train types include single motor with differential, multi-motor and in-wheel drive. HEV configurations include series, parallel, series-parallel and complex. Regenerative braking returns kinetic energy to storage.

Module 3: Energy storage includes batteries, supercapacitors, flywheels and fuel cells. Battery parameters include voltage, capacity, energy, SOC, DOD, C-rate and cycle life. BMS monitors voltage, current, temperature, SOC, SOH and balancing. EVSE supports domestic, public, fast charging and battery swapping.

Module 4: EV charging can affect distribution systems through congestion, voltage drop and power quality. Smart charging and V2X support grid operation. Renewable charging stations combine solar or wind, grid, battery buffer, energy manager and EVSE. Maintenance covers motor, drivetrain, battery, charger and recycling.

Formula Bank, Model Paper and Answers

Formula bank: Battery energy $Wh = \text{Voltage} \times Ah$. Charging time = battery energy / charger power. Charging cost = $kWh \times \text{tariff}$. Range = usable battery energy / kWh per km.

Solved example: A 400 V, 120 Ah battery has energy = $400 \times 120 = 48000 \text{ Wh} = 48 \text{ kWh}$. With a 12 kW charger, ideal charging time = $48 / 12 = 4$ hours.

Expected questions: classify EVs, compare EV and conventional vehicle, explain motor drives, draw EV subsystem diagram, explain HEV power flow, explain regenerative braking, explain BMS, explain EVSE, explain V2X and renewable charging station.

Model paper answer guide: draw block diagrams for EV configuration, drive train, battery-BMS and renewable charging station. For numerical questions, write given data, formula, substitution, answer with unit and exam tip.